



Handling Uncertainty

SE-350/SE-450, Spring-2026
Artificial Intelligence (AI)

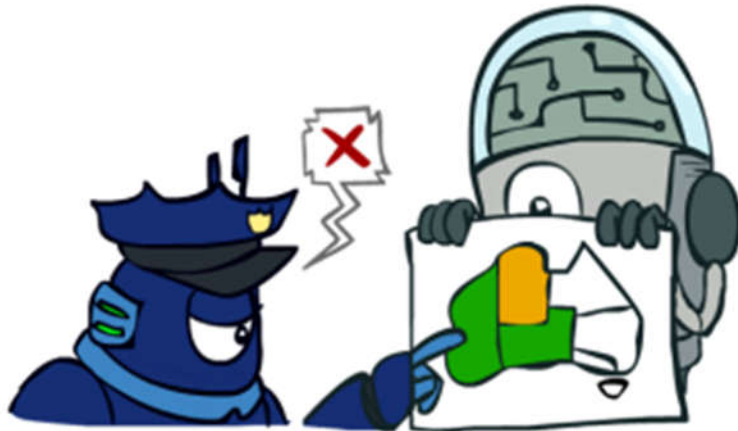
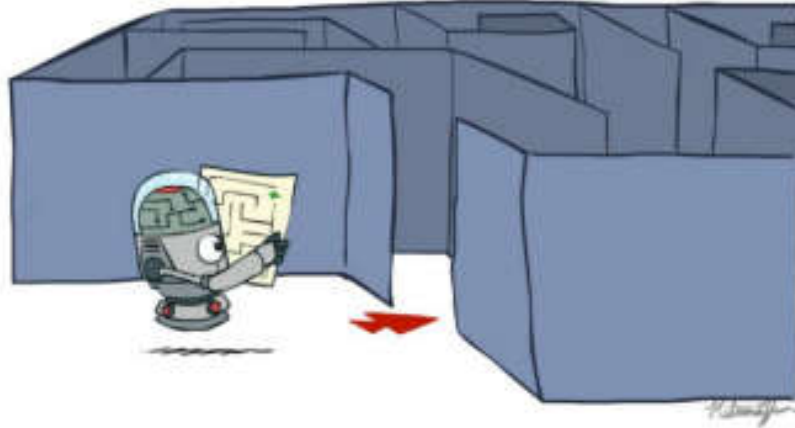
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Some slides material from
Emma Pierson & Peyrin Kao, University of California, Berkeley

Search



Constraint Satisfaction

Adversarial Search



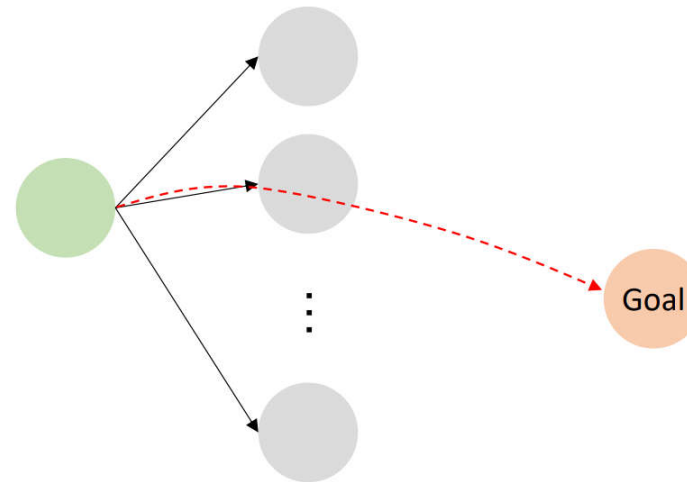
	1	2	3	4
4				Pit
3			Pit	
2				
1			Pit	

Logic

State of a game

Assignment for variables

Knowledge base



pruning, decision ordering

The techniques we learned help the computer to more efficiently search in a (exponentially) large state space.

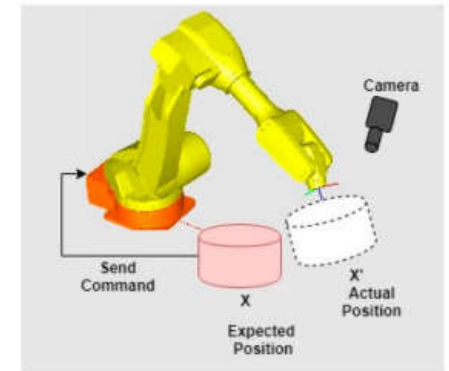
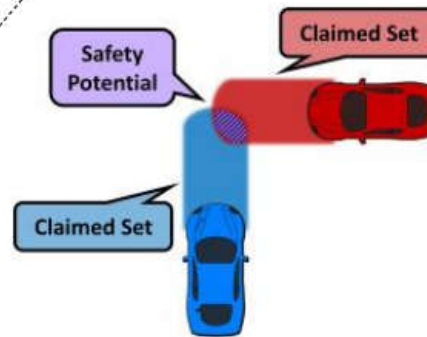
The problems we have dealt with are overall complex (large state space), but the **rules** are usually **deterministic** and **known** and **simple**.

Deterministic vs. Random/Uncertain



	1	2	3	4
4				
3				
2				
1				

Breeze $\Rightarrow$ Pit in adjacent squares



Toothache $\Rightarrow$ Cavity v Gum Problem v Abscess ..

0.3

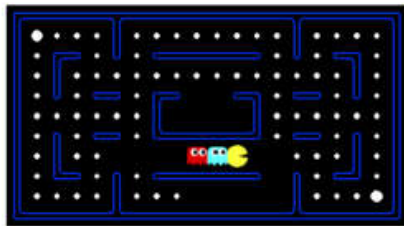
0.1

0.05

Randomness / uncertainty usually come from **ignorance**: Not all the necessary information has been or can be gathered

$\rightarrow$ Probabilistic modeling

Known vs. Unknown



Ghost takes uniformly random actions

	1	2	3	4
4				
3				
2				
1				
	Agent	Breeze	Pit	

Breeze $\Rightarrow$ Pit in adjacent squares



Toothache $\Rightarrow$ Cavity v Gum Problem v Abscess ...

0.3

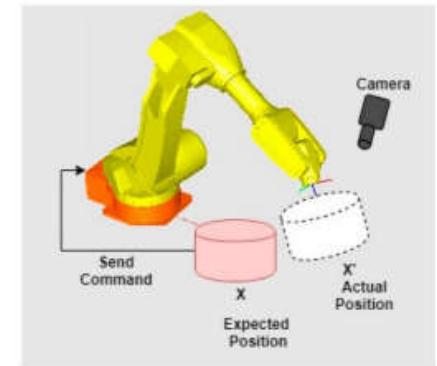
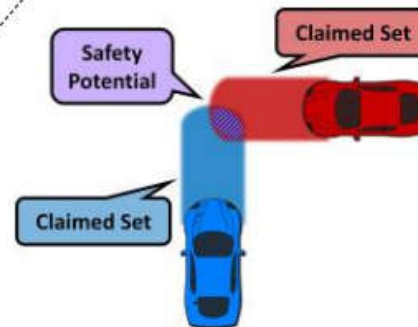
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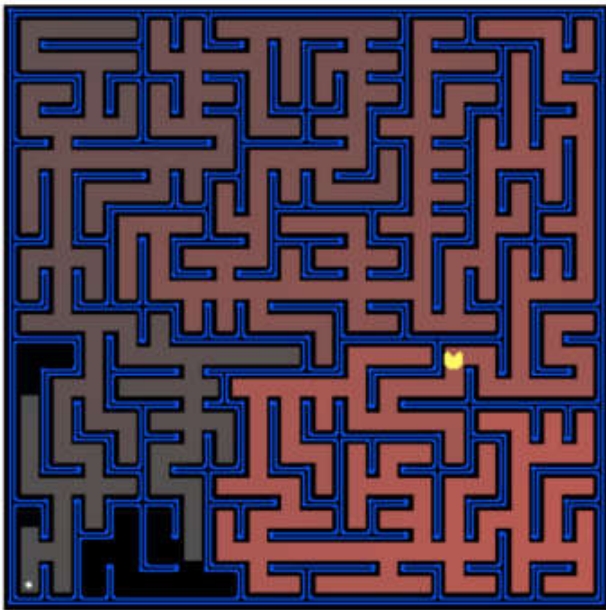
Often times, the outcome of an action or the underlying state given observations has to be **learned from experiences**

$\rightarrow$ Machine Learning

What is the **state distribution** if taking a certain action?



Simple (Easily Explainable) vs. Complicated

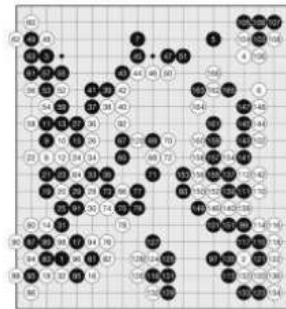


State = Pacman position

Action = NSEW

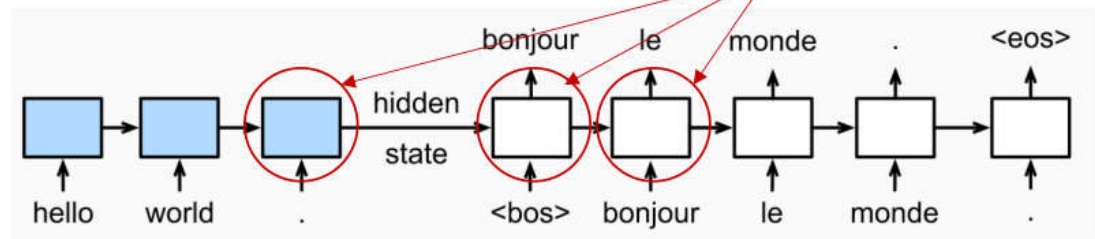
Next state = State applying Action

Heuristic = Distance to goal



Heuristic = ?

States that summarize the “meaning” of the English sentence, and the current progress of the translation



To perform the task well, we may need a good way to **encode** states (instead of its original form) and/or actions.

- Human designed features, or
- Representation Learning (Deep Learning)



Roadmap

- Search in deterministic models (finished) (Most techniques were developed before 1990)
- Probabilistic modeling
- Machine learning / deep learning: learning the model parameters and state representations from data (Most techniques were developed after 1990)
- Reinforcement learning $\approx$ performing **search and learning** simultaneously or interleavingly
- **Reminder:** this course is unable to give you a full picture of ML/DL/RL.

Probability





Uncertainty

- General situation:
 - **Observed variables (evidence):** Agent knows certain things about the state of the world (**toothache**)
 - **Unobserved variables:** Agent needs to reason about other aspects (**condition?**)
 - **Model:** Agent knows something about how the known variables relate to the unknown variables (**the probability of cavity given toothache**)
- Uncertainty modeling is a way to incorporate our beliefs and knowledge
 - Can generalize CSP and logic that we discussed before

Random Variables

- A random variable is some aspect of the world which (may) have uncertainty about

R = Is it raining?

T = Is it hot or cold?

D = How long will it take to drive to work?

L = Where is the ghost?

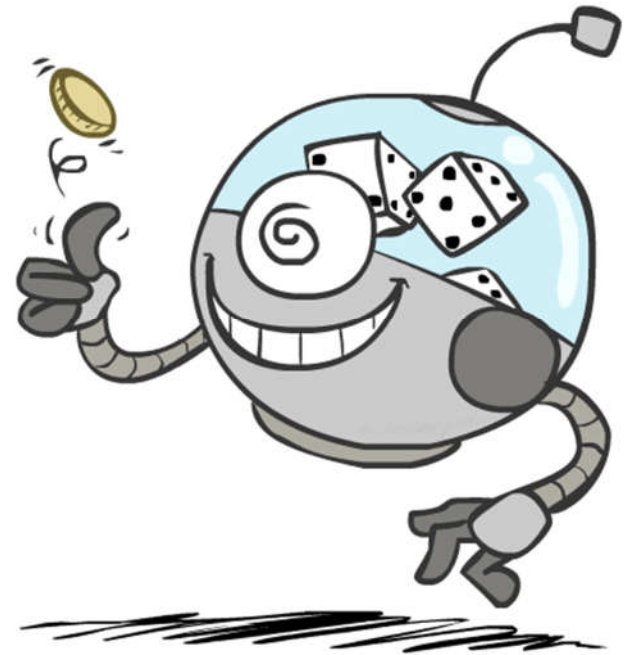
- We denote random variables with capital letters.
- Like variables in a CSP, random variables have domains

R in {true, false} (often write as {+r, -r})

T in {hot, cold}

D in $[0, \infty)$

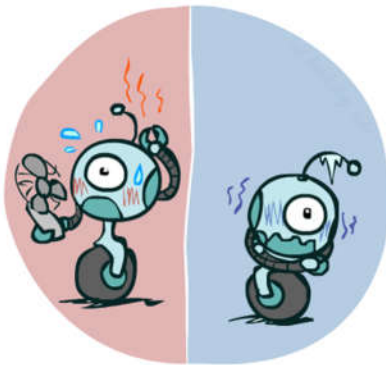
L in possible locations, maybe $\{(0,0), (0,1), \dots\}$



Probability Distributions

- Associate a probability with each value

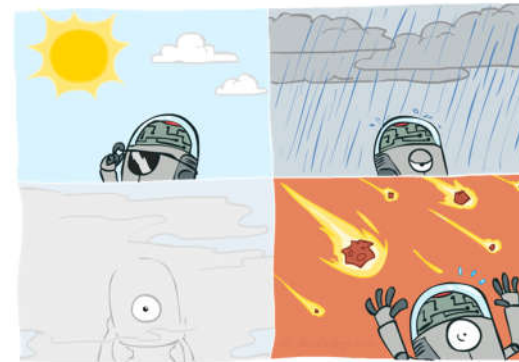
Temperature



$$P(T)$$

T	P
hot	0.5
cold	0.5

Weather



$$P(W)$$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Distributions (cont...)

- Unobserved random variables have distributions

$$P(T)$$

T	P
hot	0.5
cold	0.5

$$P(W)$$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Shorthand notation:

$$P(hot) = P(T = hot),$$

$$P(cold) = P(T = cold),$$

$$P(rain) = P(W = rain),$$

...

OK if all domain entries are unique

- A distribution is a TABLE of probabilities of values
- A probability (lower case value) is a single number

$$P(W = rain) = 0.1$$

- Must have: $\forall x \ P(X = x) \geq 0$ and $\sum_x P(X = x) = 1$

Joint Distributions

- A **joint distribution** over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or outcome):

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey: $P(x_1, x_2, \dots, x_n) \geq 0$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

- Size of distribution if n variables with domain sizes d?
 - For all but the smallest distributions, impractical to write out!

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Models

- A probabilistic model is a joint distribution over a set of random variables

- Probabilistic models:

- (Random) variables with domains
- Assignments are called outcomes
- Joint distributions: say whether assignments (outcomes) are likely
- *Normalized: sum to 1.0*
- Ideally: only certain variables directly interact

- Constraint satisfaction problems:

- Variables with domains
- Constraints: state whether assignments are possible
- Ideally: only certain variables directly interact

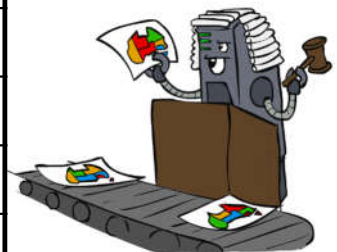
Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



Constraint over T,W

T	W	P
hot	sun	T
hot	rain	F
cold	sun	F
cold	rain	T



Events

- An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

- From a joint distribution, we can calculate the probability of any event
 - Probability that it's hot AND sunny?
 - Probability that it's hot?
 - Probability that it's hot OR sunny?

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- Typically, the events we care about are *partial assignments*, like $P(T=\text{hot})$

Quiz: Events

- $P(+x, +y)$?

$$P(X, Y)$$

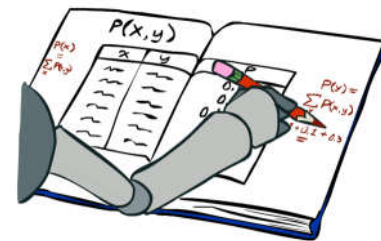
X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

- $P(+x)$?

- $P(-y \text{ OR } +x)$?

Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(t) = \sum_s P(t, s)$$

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4

$$P(s) = \sum_t P(t, s)$$

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Quiz: Marginal Distributions

$$P(X, Y)$$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

$$P(x) = \sum_y P(x, y)$$

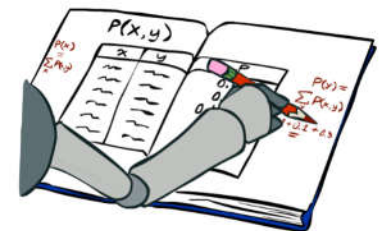
$$P(X)$$

X	P
+x	
-x	

$$P(y) = \sum_x P(x, y)$$

$$P(Y)$$

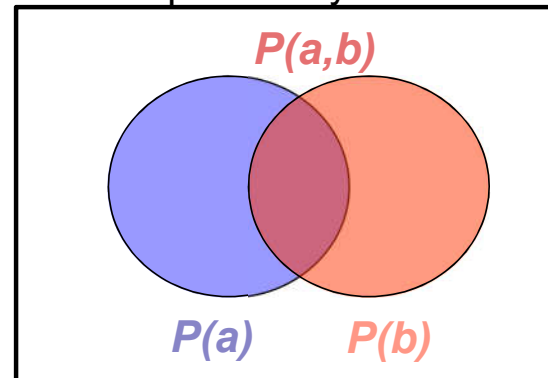
Y	P
+y	
-y	



Conditional Probabilities

- A simple relation between joint and conditional probabilities
 - In fact, this is taken as the *definition* of a conditional probability

$$P(a|b) = \frac{P(a, b)}{P(b)}$$



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$$\begin{aligned}
 &= P(W = s, T = c) + P(W = r, T = c) \\
 &= 0.2 + 0.3 = 0.5
 \end{aligned}$$

Quiz: Conditional Probabilities

- $P(+x \mid +y)$?

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

- $P(-x \mid +y)$?

- $P(-y \mid +x)$?

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$$P(W|T)$$

$P(W T = \text{hot})$	
W	P
sun	0.8
rain	0.2

$P(W T = \text{cold})$	
W	P
sun	0.4
rain	0.6

Joint Distribution

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Normalization Trick

$$\begin{aligned}
 P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\
 &= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\
 &= \frac{0.2}{0.2 + 0.3} = 0.4
 \end{aligned}$$

$P(W|T = c)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



W	P
sun	0.4
rain	0.6

$$\begin{aligned}
 P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\
 &= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\
 &= \frac{0.3}{0.2 + 0.3} = 0.6
 \end{aligned}$$

Normalization Trick (cont...)

$$\begin{aligned}
 P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\
 &= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\
 &= \frac{0.2}{0.2 + 0.3} = 0.4
 \end{aligned}$$

SELECT the joint probabilities matching the evidence

$P(c, W)$

NORMALIZE the selection (make it sum to one)

$P(W|T = c)$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



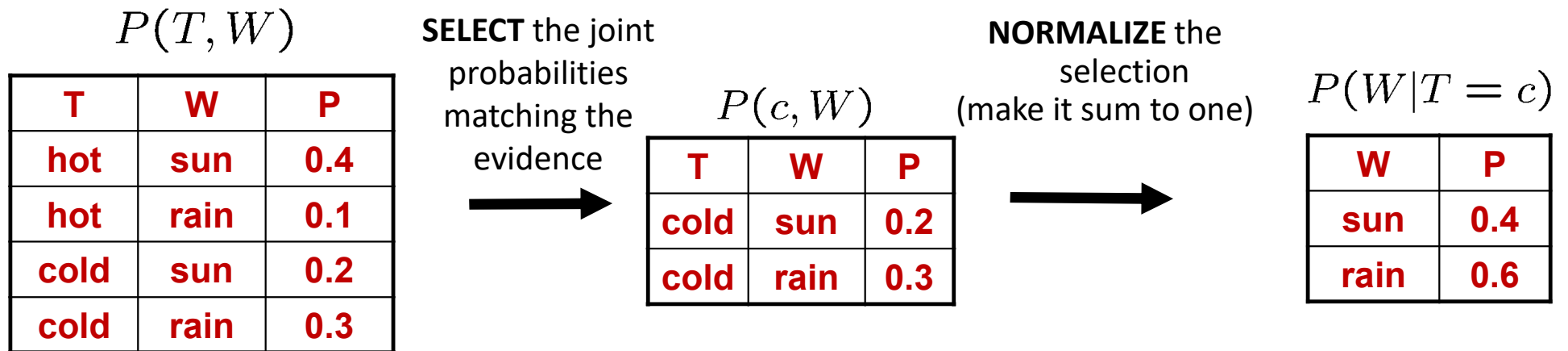
T	W	P
cold	sun	0.2
cold	rain	0.3



W	P
sun	0.4
rain	0.6

$$\begin{aligned}
 P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\
 &= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\
 &= \frac{0.3}{0.2 + 0.3} = 0.6
 \end{aligned}$$

Normalization Trick (cont...)



- Why does this work? Sum of selection is $P(\text{evidence})$! ($P(T=c)$, here)

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$

Quiz: Normalization Track

- $P(X | Y=-y)$?

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

SELECT the joint probabilities matching the evidence



NORMALIZE the selection
(make it sum to one)



To Normalize

- (Dictionary) To bring or restore to a **normal condition**

All entries sum to ONE

- Procedure:
 - Step 1: Compute $Z = \text{sum over all entries}$
 - Step 2: Divide every entry by Z

- Example 1

W	P
sun	0.2
rain	0.3

Normalize
 $Z = 0.5$

W	P
sun	0.4
rain	0.6

Example 2

T	W	P
hot	sun	20
hot	rain	5
cold	sun	10
cold	rain	15

Normalize
 $Z = 50$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Inference

- Probabilistic inference: compute a desired probability from other known probabilities (e.g. conditional from joint)
- We generally compute conditional probabilities
 - $P(\text{on time} \mid \text{no reported accidents}) = 0.90$
 - These represent the agent's *beliefs* given the evidence
- Probabilities change with new evidence:
 - $P(\text{on time} \mid \text{no accidents, 5 a.m.}) = 0.95$
 - $P(\text{on time} \mid \text{no accidents, 5 a.m., raining}) = 0.80$
 - Observing new evidence causes *beliefs to be updated*



Inference by Enumeration

- General case:
 - Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query* variable: Q
 - Hidden variables: $H_1 \dots H_r$

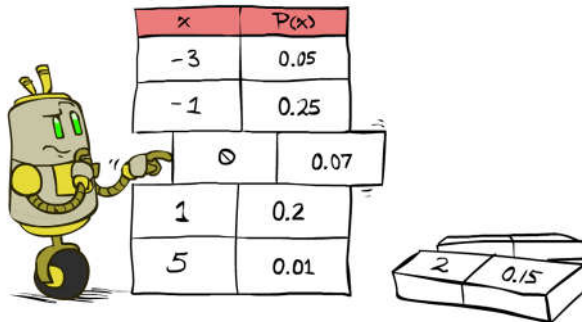
$X_1, X_2, \dots, X_n$
All variables

We want:

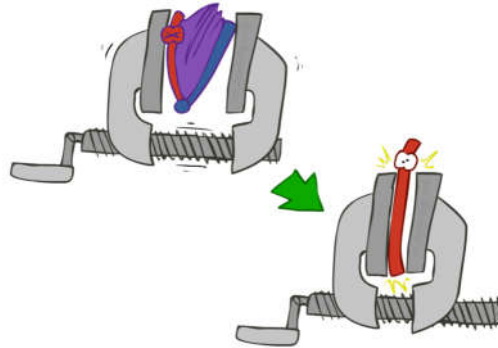
$$P(Q|e_1 \dots e_k)$$

* Works fine with multiple query variables, too

- Step 1: Select the entries consistent with the evidence



- Step 2: Sum out H to get joint of Query and evidence



$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} P(Q, \underbrace{h_1 \dots h_r}_{X_1, X_2, \dots, X_n}, e_1 \dots e_k)$$

- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

Inference by Enumeration (cont...)

- $P(W)$?
- $P(W \mid \text{winter})$?
- $P(W \mid \text{winter, hot})$?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20



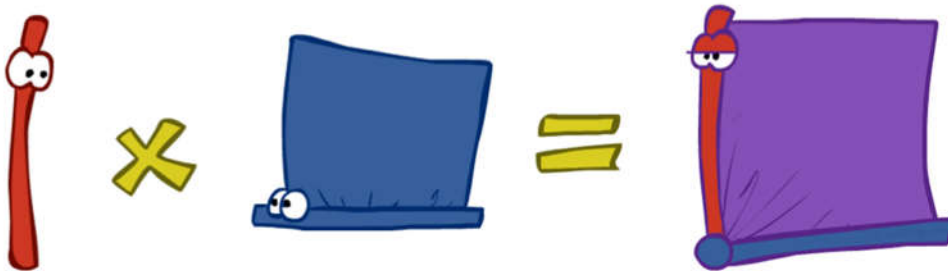
Inference by Enumeration

- **Obvious problems:**
 - Worst-case time complexity $O(d^n)$
 - Space complexity $O(d^n)$ to store the joint distribution

The Product Rule

- Sometimes have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y) \longleftrightarrow P(x|y) = \frac{P(x, y)}{P(y)}$$



The Product Rule (cont...)

$$P(y)P(x|y) = P(x, y)$$

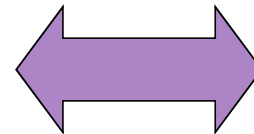
- Example

$$P(W)$$

R	P
sun	0.8
rain	0.2

$$P(D|W)$$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3



$$P(D, W)$$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	



The Chain rule

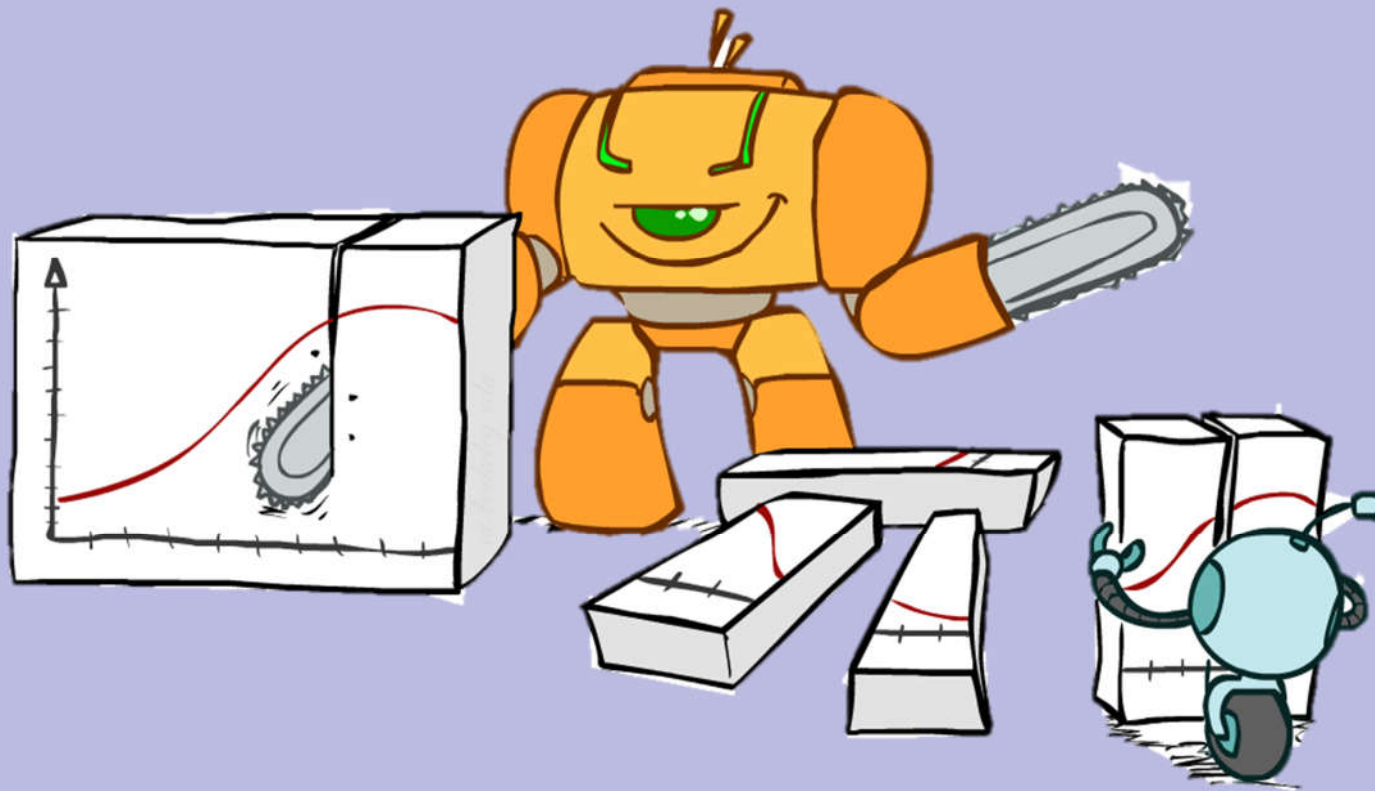
- More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

- Why is this always true?

Bayes Rule



Bayes Rule

- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Why is this at all helpful?
 - Lets us build one conditional from its reverse
 - Often one conditional is tricky but the other one is simple
 - Foundation of many systems we'll see later (e.g. ASR, MT)
- In the running for most important AI equation!

That's my rule!





Inference with Bayes' Rule

- Example: Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Example:

- M: meningitis, S: stiff neck

$$\left. \begin{array}{l} P(+m) = 0.0001 \\ P(+s|+m) = 0.8 \\ P(+s|-m) = 0.01 \end{array} \right\} \text{Example givens}$$

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{P(+s|+m)P(+m)}{P(+s|+m)P(+m) + P(+s|-m)P(-m)} = \frac{0.8 \times 0.0001}{0.8 \times 0.0001 + 0.01 \times 0.999}$$

- *Note: posterior probability of meningitis still very small*
- *Note: you should still get stiff necks checked out! Why?*



Inference with Bayes' Rule (cont...)

$$p(\textit{Prader Willi} \mid \textit{Baby weak}) = \frac{p(\textit{Baby weak} \mid \textit{Prader Willi}) p(\textit{Prader Willi})}{p(\textit{Baby weak})}$$

Quiz: Bayes' Rule

- Given:

$$P(W)$$

R	P
sun	0.8
rain	0.2

$$P(D|W)$$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

- What is $P(W | \text{dry})$?

Ghostbusters, Revisited

- Let's say we have two distributions:
 - Prior distribution over ghost location: $P(G)$
 - Let's say this is uniform

Sensor reading model: $P(R | G)$

Given: we know what our sensors do

R = reading color measured at $(1,1)$

E.g. $P(R = \text{yellow} | G=(1,1)) = 0.1$

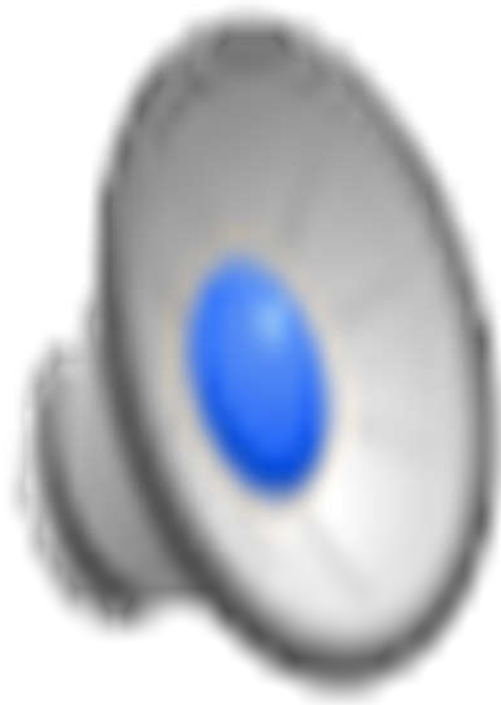
- We can calculate the **posterior distribution** $P(G|r)$ over ghost locations given a reading using Bayes' rule:

$$P(g|r) \propto P(r|g)P(g)$$

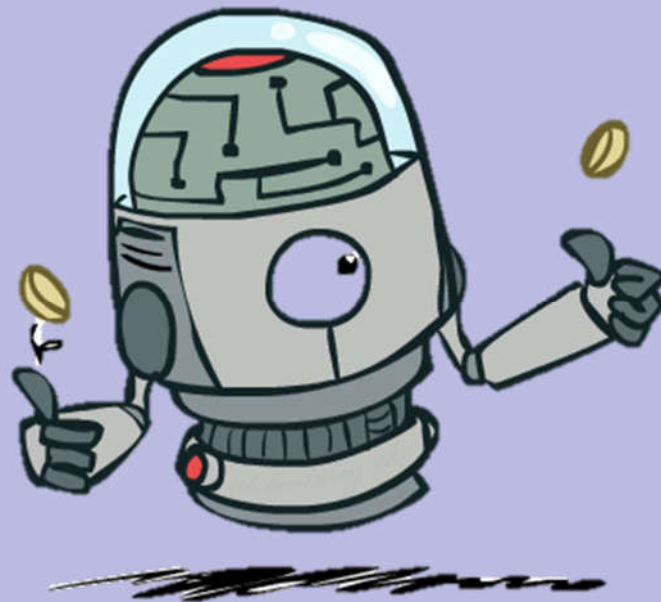
0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

0.17	0.10	0.10
0.09	0.17	0.10
<0.01	0.09	0.17

Video of Demo Ghostbusters with Probability



Independence

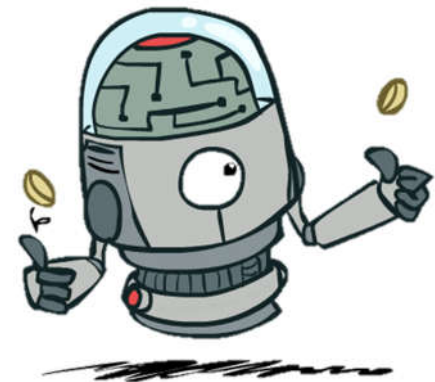


Independence

- Two variables are independent if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

- This says that their joint distribution factors into a product two simpler distributions
- Another form: $\forall x, y : P(x|y) = P(x)$
- We write: $X \perp\!\!\!\perp Y$
- Independence is a simplifying modeling assumption
 - Empirical joint distributions: at best “close” to independent
 - What could we assume for {Weather, Traffic, Cavity, Toothache}?



Example: Independence?

$$P(T)$$

T	P
hot	0.5
cold	0.5

$$P_1(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W)$$

W	P
sun	0.6
rain	0.4

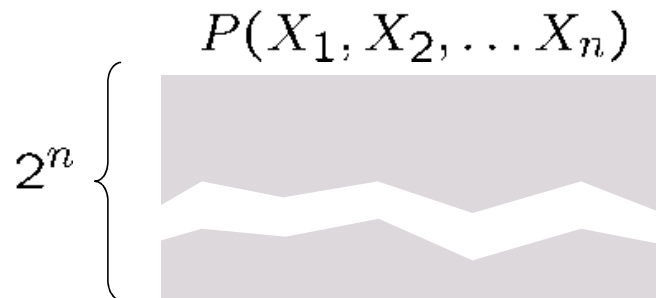
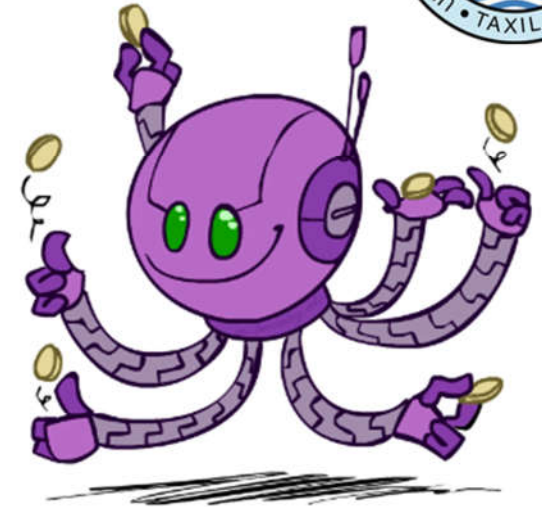
$$P_2(T, W)$$

T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

Example: Independence

- N fair, independent coin flips:

$P(X_1)$		$P(X_2)$		$\dots$		$P(X_n)$	
H	0.5	H	0.5			H	0.5
T	0.5	T	0.5			T	0.5



Conditional Independence

- **P(Toothache, Cavity, Catch)**

- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:

- $P(+catch \mid +toothache, +cavity) = P(+catch \mid +cavity)$

- The same independence holds if I don't have a cavity:

- $P(+catch \mid +toothache, -cavity) = P(+catch \mid -cavity)$

- Catch is **conditionally independent** of Toothache given Cavity:

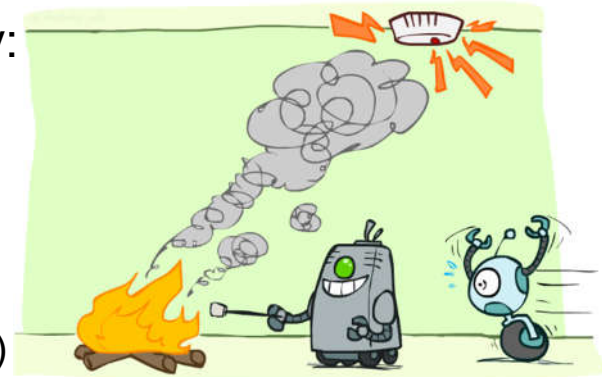
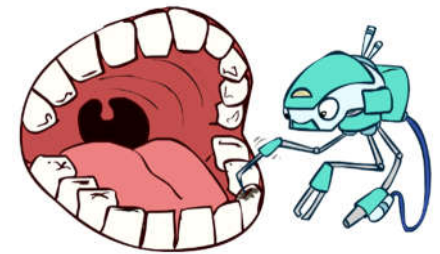
- $P(Catch \mid Toothache, Cavity) = P(Catch \mid Cavity)$

- **Equivalent statements:**

- $P(Toothache \mid Catch, Cavity) = P(Toothache \mid Cavity)$

- $P(Toothache, Catch \mid Cavity) = P(Toothache \mid Cavity) P(Catch \mid Cavity)$

- One can be derived from the other easily



Conditional Independence (cont...)

- Unconditional (absolute) independence very rare (why?)
- Conditional independence is our most basic and robust form of knowledge about uncertain environments.
- X is conditionally independent of Y given Z $X \perp\!\!\!\perp Y | Z$

if and only if:

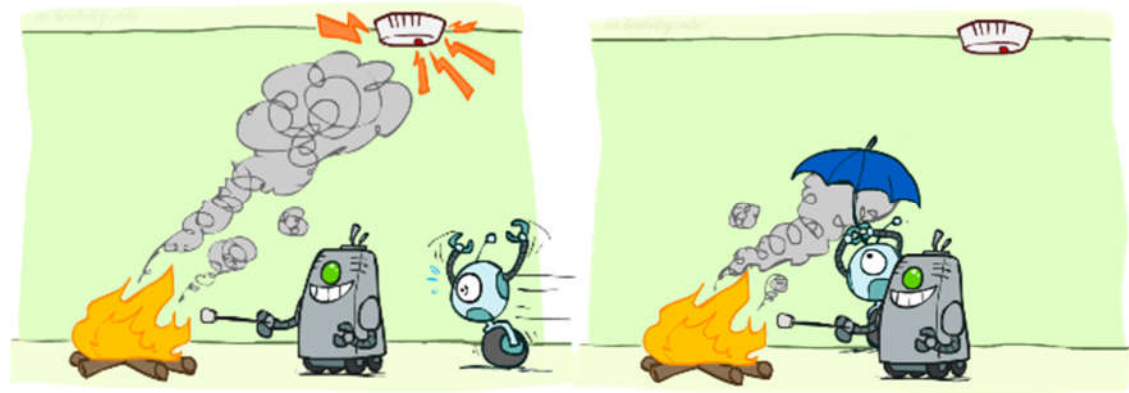
$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

or, equivalently, if and only if

$$\forall x, y, z : P(x|z, y) = P(x|z)$$

Conditional Independence (cont...)

- What about this domain:
 - Traffic
 - Umbrella
 - Raining
- What about this domain:
 - Fire
 - Smoke
 - Alarm





Conditional Independence and the Chain Rule

- Chain rule: $P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots$

- Trivial decomposition:

$$P(\text{Traffic, Rain, Umbrella}) =$$

$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain, Traffic})$$

- With assumption of conditional independence:

$$P(\text{Traffic, Rain, Umbrella}) =$$

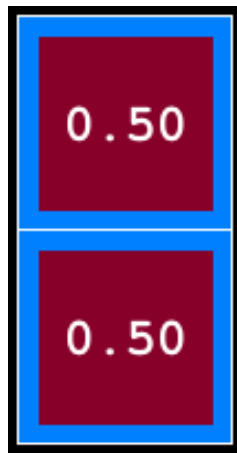
$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

- Bayes'nets / graphical models help us express conditional independence assumptions

Ghostbusters Chain Rule

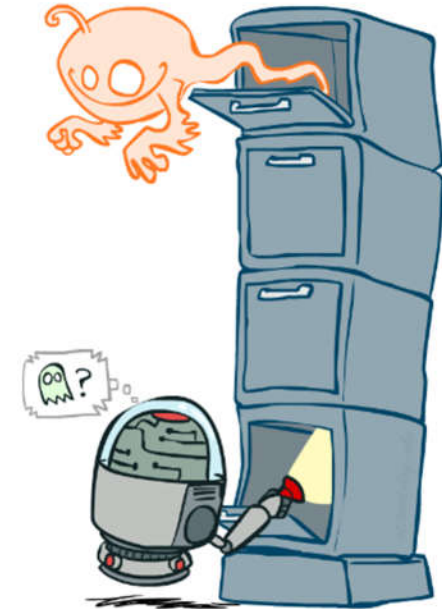
- Each sensor depends only on where the ghost is
- That means, the two sensors are conditionally independent, given the ghost position
- T: Top square is red
B: Bottom square is red
G: Ghost is in the top

- Givens:
 $P(+g) = 0.5$
 $P(-g) = 0.5$
 $P(+t | +g) = 0.8$
 $P(+t | -g) = 0.4$
 $P(+b | +g) = 0.4$
 $P(+b | -g) = 0.8$



$$P(T,B,G) = P(G) P(T|G) P(B|G)$$

T	B	G	P(T,B,G)
+t	+b	+g	0.16
+t	+b	-g	0.16
+t	-b	+g	0.24
+t	-b	-g	0.04
-t	+b	+g	0.04
-t	+b	-g	0.24
-t	-b	+g	0.06
-t	-b	-g	0.06





Topics Covered

- *Artificial Intelligence: A Modern Approach (4th Edition) by Stuart Russell and Peter Norvig. (Chapter -12 (12.1-12.5))*

Any Question?